

## 7.1 Multimodal Learning

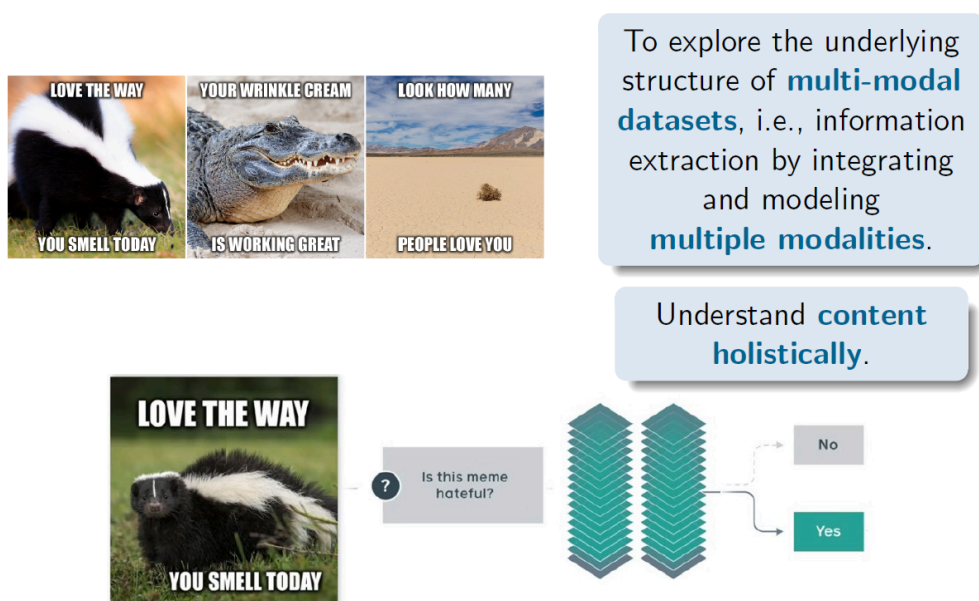
Two or more views of the same objects.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

Figures and notes follow Baltrušaitis, Ahuja, and Morency, *Multimodal Machine Learning: A Survey and Taxonomy*.

### 1. Why multimodal learning

Several **modalities** at once — language, vision, audio — rather than one stream. Complementary signal, and redundant signal. The rest of the note is a taxonomy of the hard parts.



### 2. Challenges in multimodal fusion

**Representation.** Learn a summary that uses complementarity and redundancy. Heterogeneity makes this hard. Language is symbolic; audio and vision are signals.

**Translation.** Map one modality to another. Heterogeneity again, and the map can be open-ended or subjective. Many correct captions for one image; no single perfect translation.

**Alignment.** Direct relations between *elements* of different modalities. You need a similarity, long-range dependence, and a way to resolve ambiguity. Example: recipe steps lined up with a cooking video.

**Fusion.** Combine modalities for a prediction. Example: audio-visual speech recognition (lip motion plus the speech signal). Predictive power can differ by modality; noise topology can differ; data can be

missing.

**Co-learning.** Transfer knowledge across modalities, representations, and predictors. Co-training, conceptual grounding, zero-shot learning. Useful when one modality is resource-poor (few annotations).

### 3. Multimodal representations

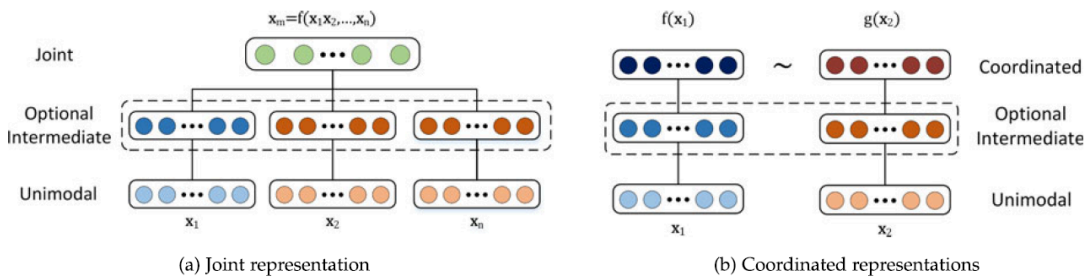
A computational model needs a **representation**. Multimodal representation uses information from several entities, each a vector or a tensor.

Hard parts: combining heterogeneous data, different noise levels, missing data. Representation quality shows up in speech recognition and object classification.

Wanted properties in general: smoothness, coherence, sparsity, natural clustering. Extra for multimodal: similarity between concepts is reflected, representations are easy to obtain, missing modalities can be handled.

Two types:

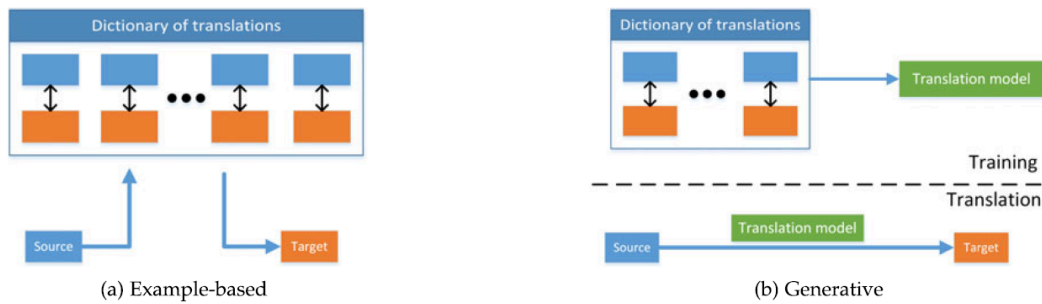
| Type        | What it does                                                                              |
|-------------|-------------------------------------------------------------------------------------------|
| Joint       | Combine modalities into <b>one</b> space. A function of the unimodal representations.     |
| Coordinated | Process modalities <b>separately</b> , then enforce similarity or structure between them. |



### 4. Translation

Generate text from images, or the reverse.

- **Example-based.** A dictionary. Retrieval finds the closest match. Combination-based models mix examples. **Generative** models build a map and emit a sequence of symbols — harder.
- **Grammar-based.** Templates and a restricted grammar.
- **Encoder-decoder.** Encode the source modality, decode the target.
- **Continuous generation.** An output at each time step. Natural for sequence-to-sequence.



## 5. Multimodal alignment

Relations between **sub-components** of instances from different modalities.

**Explicit.** Align those sub-components on purpose. Unsupervised: DTW, graphical models. Supervised: labeled pairs. Deep: CNNs, LSTMs.

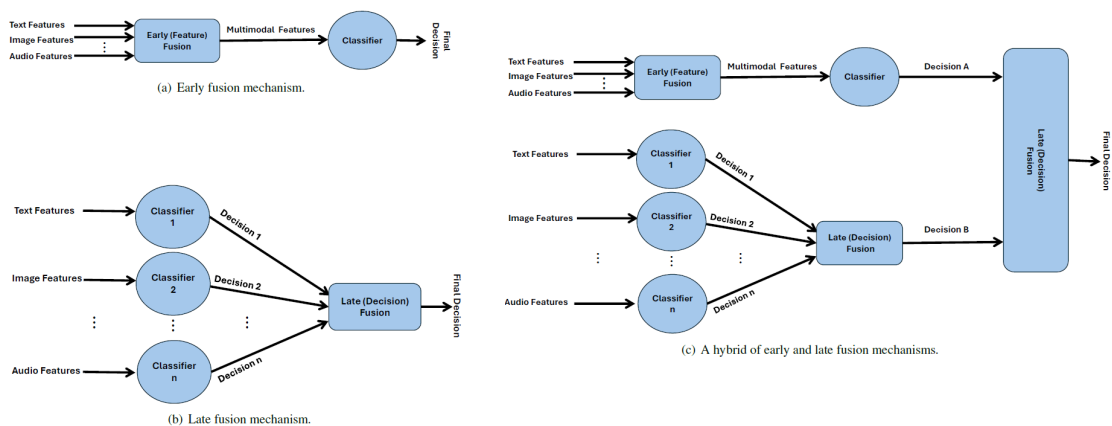
**Implicit.** Alignment as an intermediate for another task. Early work used graphical models. Modern nets use **attention**.

Applications: multimedia retrieval, translation, question answering.

## 6. Multimodal fusion

Model-agnostic types:

| Fusion        | When you combine                             |
|---------------|----------------------------------------------|
| <b>Early</b>  | Right after feature extraction               |
| <b>Late</b>   | After each modality has made a decision      |
| <b>Hybrid</b> | Early-fusion output plus unimodal predictors |

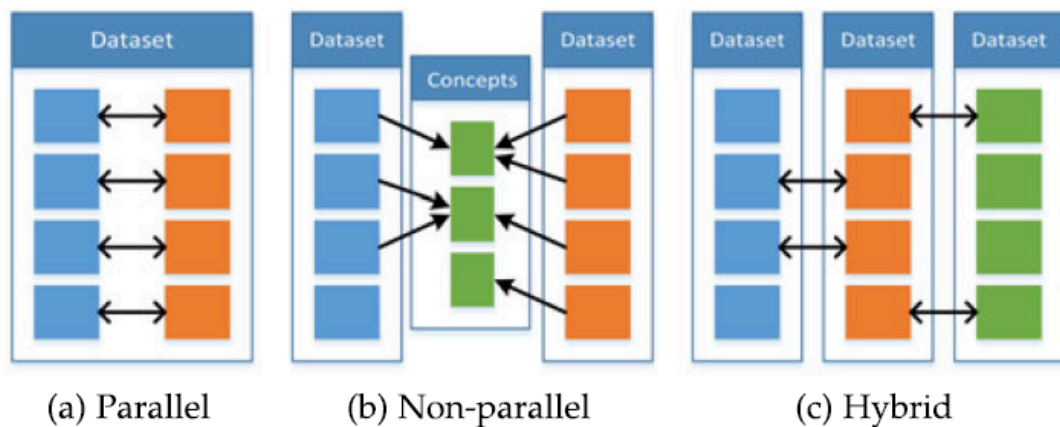


## 7. Co-learning

Improve a **resource-poor** modality with a **resource-rich** one (annotations, reliable labels).

| Approach     | Link between modalities                   |
|--------------|-------------------------------------------|
| Parallel     | Directly linked observations              |
| Non-parallel | No direct links; often shared categories  |
| Hybrid       | A shared modality or data set as a bridge |

The helper modality is typically used in **training only**, not at test time.



## Practice

1. Name the five challenges in the taxonomy, and give one example of **alignment**.
2. What is the difference between a **joint** representation and a **coordinated** one?
3. When would **late** fusion be a better default than **early** fusion?